

JEONG-SOO LEE

其他

基本信息

姓名: JEONG-SOO LEE

所属院校: 浦项理工大学

地区: South Korea

学科领域

微电子科学与工程

材料物理

集成电路设计与集成系统

电子科学与技术

生物医学工程

纳米材料与技术

研究领域

纳米电子与先进工艺

微电子器件与制造

半导体与光电子材料

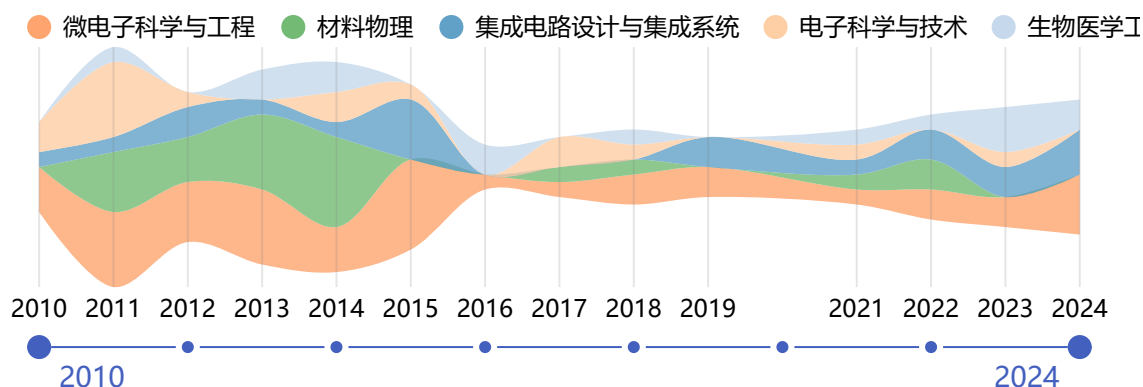
生物医学传感与检测技术

半导体器件物理与工艺

个人简介

JEONG - SOO LEE is a teacher in the Department of Electrical Engineering at the Pohang University of Science and Technology. His academic citation statistics are remarkable, with a total of 3056 bibliographic records, an h - index of 28, and an i10 - index of 65. Since 2020, he has 972 bibliographic records, an h - index of 15, and an i10 - index of 22. He has published many influential academic papers, such as "Chemical synthesis of PEDOT nanotubes" in 2006 with 250 citations and "Isolation method for semiconductor device" in 2002 with 228 citations. His research interests include semiconductors, sensors, and nanotechnology.

近年发表的论文



Sensing Characteristics of SARS-CoV-2 Spike Protein Using Aptamer-Functionalized Si-Based Electrolyte-Gated Field-Effect Transistor (EGT)

BIOSENSORS-BASEL

Shin, Seonghwan; Kim, Sangwon; Choi, Wonyeong; Do, Jeonghyeon; Son, Jongmin; Kim, Kihyun; Jang, Sungkey; Lee, Jeong-Soo

被引用次数: 1

2024

Influence of Surface Treatments on Urea Detection Using Si Electrolyte-Gated Transistors with Different Gate Electrodes

MICROMACHINES

Choi, Wonyeong; Shin, Seonghwan; Do, Jeonghyeon; Son, Jongmin; Kim, Kihyun; Lee, Jeong-Soo

被引用次数: 0

2024

Effect of Quasi-One-Dimensional Properties on Source/Drain Contacts in Vertical Nanowire Field-Effect Transistors (VNWFETs)

MICROMACHINES

Park, Iksoo; Choi, Jaeyong; Kim, Jungsik; Kong, Byoung Don; Lee, Jeong-Soo

被引用次数: 0

2024

Impact of Electron and Hole Trap Profiles in BE-TOX on Retention Characteristics of 3D NAND Flash Memory

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Yoon, Gilsang; Go, Donghyun; Park, Jounghun; Kim, Donghwi; Kim, Jongwoo; An, Ukju; Kim, Jungsik; Lee, Jeong-Soo; Kong, Byoung Don

被引用次数: 0

2024

Novel Dummy Cell Programming Scheme to Improve Retention Characteristics in 3-D NAND Flash Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Donghwi; Yoon, Gilsang; Go, Donghyun; Park, Jounghun; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 0

2024

Decomposition of Charge Loss Mechanisms in 3-D Nand Flash Memory: Impact of Cell Dimension via High-Temperature Long-Term Retention Characteristics

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Jounghun; Yoon, Gilsang; Go, Donghyun; Kim, Donghwi; Sagong, Hyun Chul; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 0

2024

Effects of Poly-Si Grain Boundary on Retention Characteristics under Cross-Temperature Conditions in 3-D NAND Flash Memory

MICROMACHINES

An, Ukju; Yoon, Gilsang; Go, Donghyun; Park, Jounghun; Kim, Donghwi; Kim, Jongwoo; Lee, Jeong-Soo

被引用次数: 1

2023

One-step immunoassay based on filtration for detection of food poisoning-related bacteria

TALANTA

Kim, Hong-Rae; Bong, Ji-Hong; Kang, Min-Jung; Shim, Won-Bo; Lee, Jeong-Soo; Pyun, Jae-Chul

被引用次数: 7

2023

Hydrogel-Gated Silicon Nanotransistors for SARS-CoV-2 Antigen Detection in Physiological Ionic Strength

ADVANCED MATERIALS INTERFACES

Parichenko, Alexandra; Choi, Wonyeong; Shin, Seonghwan; Schlecht, Marlena; Gutierrez, Rafael; Akbar, Teuku Fawzul; Werner, Carsten; Lee, Jeong-Soo; Ibarlucea, Bergoi; Cuniberti, Gianaurelio

被引用次数: 4

2023

Effect of Noncircular Channel on Distribution of Threshold Voltage in 3D NAND Flash Memory

MICROMACHINES

Go, Donghyun; Yoon, Gilsang; Park, Jounghun; Kim, Donghwi; Kim, Jiwon; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 0

2023

Sybodies as Novel Bioreceptors toward Field-Effect Transistor-Based Detection of SARS-CoV-2 Antigens

ACS APPLIED MATERIALS & INTERFACES

Zhang, Chi; Parichenko, Alexandra; Choi, Wonyeong; Shin, Seonghwan; Panes-Ruiz, Luis Antonio; Belyaev, Dmitry; Custodio, Tania Filipa; Low, Christian; Lee, Jeong-Soo; Ibarlucea, Bergoi; Cuniberti, Gianaurelio

被引用次数: 1

2023

Highly Sensitive Detection of Urea Using Si Electrolyte-Gated Transistor with Low Power Consumption

BIOSENSORS-BASEL

Choi, Wonyeong; Jin, Bo; Shin, Seonghwan; Do, Jeonghyeon; Son, Jongmin; Kim, Kihyun; Lee, Jeong-Soo

被引用次数: 1

2023

Single-Event Upset in 3-D Charge-Trap NAND Flash Memories

IEEE TRANSACTIONS ON ELECTRON DEVICES

Park, Jounghun; Han, Jin-Woo; Yoon, Gilsang; Go, Donghyun; Kim, Donghwi; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 1

2022

Improvement of Fermi-Level Pinning and Contact Resistivity in Ti/Ge Contact Using Carbon Implantation

MICROMACHINES

Park, Iksoo; Lee, Donghun; Jin, Bo; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 3

2022

An Ultrasensitive Silicon-Based Electrolyte-Gated Transistor for the Detection of Peanut Allergens

BIOSENSORS-BASEL

Kim, Donghoon; Jin, Bo; Kim, Sol-A; Choi, Wonyeong; Shin, Seonghwan; Park, Jiwon; Shim, Won-Bo; Kim, Kihyun; Lee, Jeong-Soo

被引用次数: 5

2022

Effects of Carbon Incorporation on Electrical Characteristics and Thermal Stability of Ti/TiO₂/n-Ge MIS Contact

IEEE ACCESS

Park, Iksoo; Shin, Seonghwan; Kim, Jungsik; Jin, Bo; Lee, Jeong-Soo

被引用次数: 0

2022

Impact of P/E Stress on Trap Profiles in Bandgap-Engineered Tunneling Oxide of 3D NAND Flash Memory

IEEE ACCESS

Yoon, Gilsang; Ko, Donghyun; Park, Jounghun; Kim, Donghwi; Kim, Jungsik; Lee, Jeong-Soo

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2022

Fabrication and Characterization of Nanonet-Channel LTPS TFTs Using a Nanosphere-Assisted Patterning Technique

MICROMACHINES

Yoon, Gilsang; Kim, Donghoon; Park, Iksoo; Jin, Bo; Lee, Jeong-Soo

被引用次数: 3

2021

Three-dimensional imaging of carbon clusters in thermally stable nickel silicides by carbon pre-implantation

APPLIED SURFACE SCIENCE

Park, Iksoo; Seol, Jae Bok; Yoon, Gilsang; Lee, Jeong-Soo

被引用次数: 3

2021

Lumped-Capacitive Modeling and Sensing Characteristics of an Electrolyte-Gated FET Biosensor for the Detection of the Peanut Allergen

IEEE ACCESS

Kim, Donghoon; Choi, Wonyeong; Shin, Seonghwan; Park, Jiwon; Kim, Kihyun; Jin, Bo; Lee, Jeong-Soo

被引用次数: 1

2021

Extraction of Nitride Trap Profile in 3-D NAND Flash Memory Using Intercell Program Pattern

IEEE ACCESS

Park, Jounghun; Yoon, Gilsang; Go, Donghyun; Kim, Jungsik; Lee, Jeong-Soo

被引用次数: 7

2021

Improved Long-Term Responses of Au-Decorated Si Nanowire FET Sensor for NH₃ Detection

IEEE SENSORS JOURNAL

Kim, Donghoon; Park, Chanoh; Choi, Wonyeong; Shin, Seong-Hwan; Jin, Bo; Baek, Rock-Hyun; Lee, Jeong-Soo

被引用次数: 30

2020

Caution: Abnormal Variability Due to Terrestrial Cosmic Rays in Scaled-Down FinFETs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jungsik; Lee, Jeong-Soo; Han, Jin-Woo; Meyyappan, M.

被引用次数: 20

2019

Soft Error in Saddle Fin Based DRAM

IEEE ELECTRON DEVICE LETTERS

Han, Jin-Woo; Kim, Jungsik; Moon, Dong-Il; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 17

2019

Electrical Characteristics and pH Response of a Parylene-H Sensing Membrane in a Si-Nanonet Ion-Sensitive Field-Effect Transistor

SENSORS

Jin, Bo; Lee, Ga-Yeon; Park, ChanOh; Kim, Donghoon; Choi, Wonyeong; Yoo, Jae-Woo; Pyun, Jae-Chul; Lee, Jeong-Soo

被引用次数: 9

2018

Single-Event Transient in FinFETs and Nanosheet FETs

IEEE ELECTRON DEVICE LETTERS

Kim, Jungsik; Lee, Jeong-Soo; Han, Jin-Woo; Meyyappan, M.

被引用次数: 50

2018

Systematic analysis of oxide trap distribution of 4H-SiC DMOSFETs using TSCIS and its correlation with BTI and SILC behavior

SOLID-STATE ELECTRONICS

Baek, Sangwon; Lee, Junyoung; Park, Iksoo; Baek, Rock-Hyun; Lee, Jeong-Soo

被引用次数: 1

2018

Threshold voltage variation depending on single grain boundary and stored charges in an adjacent cell for vertical silicon-oxide-nitride-oxide-silicon NAND flash memory

JAPANESE JOURNAL OF APPLIED PHYSICS

Oh, Hyeongwan; Kim, Jiwon; Baek, Rock-Hyun; Lee, Jeong-Soo

被引用次数: 8

2018

Impact of geometrical parameters on the electrical performance of network-channel polycrystalline silicon thin-film transistors

JAPANESE JOURNAL OF APPLIED PHYSICS

Lee, Junyoung; Lee, Hojoon; Jin, Bo; Oh, Hyeongwan; Baek, Sangwon; Yoon, Gilsang; Lee, Yongsu; Baek, Rock-Hyun; Lee, Jeong-Soo

被引用次数: 2

2018

Highly Enhanced Performance of Network Channel Polysilicon Thin-Film Transistors

IEEE ELECTRON DEVICE LETTERS

Lee, Hojoon; Lee, Junyoung; Baek, Sangwon; Jeong, Woong Hee; Lee, Yongsu; Yang, Taehoon; Lee, Jeong-Soo

被引用次数: 5

2017

Electrical characteristics of tunneling field-effect transistors with asymmetric channel thickness

JAPANESE JOURNAL OF APPLIED PHYSICS

Kim, Jungsik; Oh, Hyeongwan; Kim, Jiwon; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 3

2017

Buffer effects of two functional groups against pH variation at aminosilanized Electrolyte-Oxide-Semiconductor (EOS) capacitor

SENSORS AND ACTUATORS B-CHEMICAL

Hong, Nanki; Park, ChanOh; Kim, Donghoon; Jeong, Ki-Soo; Yoon, Jun-Sik; Jin, Bo; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 4

2017

Work function consideration in vacuum field emission transistor design

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Kim, Jiwon; Oh, Hyeongwan; Kim, Jungsik; Baek, Rock-Hyun; Han, Jin-Woo; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 11

2017

A Reconfigurable and Portable Highly Sensitive Biosensor Platform for ISFET and Enzyme-Based Sensors

IEEE SENSORS JOURNAL

Lee, Ikho; Lee, Seung-Woo; Lee, Ki-Young; Park, Chanoh; Kim, Donghoon; Lee, Jeong-Soo; Yi, Hyunjung; Kim, Byungsub

被引用次数: 13

2016

Silicon nanowire biosensors for detection of cardiac troponin I (cTnI) with high sensitivity

BIOSENSORS & BIOELECTRONICS

Kim, Kihyun; Park, Chanoh; Kwon, Donghoon; Kim, Donghoon; Meyyappan, M.; Jeon, Sangmin; Lee, Jeong-Soo

被引用次数: 163

2016

Effects of single grain boundary and random interface traps on electrical variations of sub-30 nm polysilicon nanowire structures

MICROELECTRONIC ENGINEERING

Oh, Hyeongwan; Kim, Jungsik; Lee, Junyoung; Rim, Taiuk; Baek, Chang-Ki; Lee, Jeong-Soo

被引用次数: 13

2016

Design guidelines for nanoscale vacuum field emission transistors

JOURNAL OF VACUUM SCIENCE & TECHNOLOGY B

Kim, Jungsik; Kim, Jiwon; Oh, Hyeongwan; Meyyappan, M.; Han, Jin-Woo; Lee, Jeong-Soo

被引用次数: 33

2016

Investigation of RC Parasitics Considering Middle-of-the-Line in Si-Bulk FinFETs for Sub-14-nm Node Logic Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Jeong, Eui-Young; Yoon, Jun-Sik; Baek, Chang-Ki; Kim, Ye-Ram; Hong, Jae-Ho; Lee, Jeong-Soo; Baek, Rock-Hyun; Jeong, Yoon-Ha

被引用次数: 15

2015

Junction Design Strategy for Si Bulk FinFETs for System-on-Chip Applications Down to the 7-nm Node

IEEE ELECTRON DEVICE LETTERS

Yoon, Jun-Sik; Jeong, Eui-Young; Baek, Chang-Ki; Kim, Ye-Ram; Hong, Jae-Ho; Lee, Jeong-Soo; Baek, Rock-Hyun; Jeong, Yoon-Ha

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2015

Impact of the spacer dielectric constant on parasitic RC and design guidelines to optimize DC/AC performance in 10-nm-node Si-nanowire FETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Hong, Jae-Ho; Lee, Sang-Hyun; Kim, Ye-Ram; Jeong, Eui-Young; Yoon, Jun-Sik; Lee, Jeong-Soo; Baek, Rock-Hyun; Jeong, Yoon-Ha

被引用次数: 11

2015

Physical DC and thermal noise models of 18nm double-gate junctionless p-type MOSFETs for low noise RF applications

JAPANESE JOURNAL OF APPLIED PHYSICS

Jeong, Eui-Young; Deen, M. Jamal; Chen, Chih-Hung; Baek, Rock-Hyun; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 1

2015

Extraction of source/drain resistivity parameters optimized for double-gate FinFETs

JAPANESE JOURNAL OF APPLIED PHYSICS

Yoon, Jun-Sik; Jeong, Eui-Young; Lee, Sang-Hyun; Kim, Ye-Ram; Hong, Jae-Ho; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 4

2015

Three-dimensional simulation of threshold voltage variations due to an oblique single grain boundary in sub-40nm polysilicon nanowire FETs

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Kim, Jungsik; Oh, Hyeongwan; Lee, Junyoung; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 5

2015

Threshold Voltage Variations Due to Oblique Single Grain Boundary in Sub-50-nm Polysilicon Channel

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Jungsik; Rim, Taiuk; Lee, Junyoung; Baek, Chang-Ki; Meyyappan, Meyya; Lee, Jeong-Soo

被引用次数: 12

2014

Investigation of Low-Frequency Noise in p-type Nanowire FETs: Effect of Switched Biasing Condition and Embedded SiGe Layer

IEEE ELECTRON DEVICE LETTERS

Lee, Sang-Hyun; Kim, Ye-Ram; Hong, Jae-Ho; Jeong, Eui-Young; Yoon, Jun-Sik; Baek, Chang-Ki; Kim, Dong-Won; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 4

2014

Silicon Nanowire Biologically Sensitive Field Effect Transistors: Electrical Characteristics and Applications

JOURNAL OF NANOSCIENCE AND NANOTECHNOLOGY

Rim, Taiuk; Baek, Chang-Ki; Kim, Kihyun; Jeong, Yoon-Ha; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 15

2014

Synthesis and characterization of carbon nanowalls on different substrates by radio frequency plasma enhanced chemical vapor deposition

CARBON

Davami, Keivan; Shaygan, Mehrdad; Kheirabi, Nazli; Zhao, Jiong; Kovalenko, Dana A.; Ruemmeli, Mark H.; Opitz, Joerg; Cuniberti, Gianaurelio; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 127

2014

Thermally Phase-Transformed In₂Se₃ Nanowires for Highly Sensitive Photodetectors

SMALL

Kang, Daegun; Rim, Taiuk; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 30

2014

Suspended honeycomb nanowire ISFETs for improved stiction-free performance

NANOTECHNOLOGY

Kim, Kihyun; Rim, Taiuk; Park, Chanoh; Kim, Donghoon; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 19

2014

Ga-doped indium oxide nanowire phase change random access memory cells

NANOTECHNOLOGY

Jin, Bo; Lim, Taekyung; Ju, Sanghyun; Latypov, Marat I.; Kim, Hyoung Seop; Meyyappan, M.; Lee, Jeong-Soo

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2014

Investigation of thermal resistance and power consumption in Ga-doped indium oxide (In₂O₃) nanowire phase change random access memory

APPLIED PHYSICS LETTERS

Jin, Bo; Lim, Taekyung; Ju, Sanghyun; Latypov, Marat I.; Pi, Dong-Hai; Kim, Hyoung Seop; Meyyappan, M.; Lee, Jeong-Soo

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2014

Annealing effect on the thermal conductivity of thermoelectric ZnTe nanowires

MATERIALS LETTERS

Shaygan, Mehrdad; Kheirabi, Nazli; Davami, Keivan; Mortazavi, Bohayra; Lee, Jeong-Soo; Cuniberti, Gianaurelio; Meyyappan, M.

被引用次数: 3

2014

Role of an encapsulating layer for reducing resistance drift in phase change random access memory

AIP ADVANCES

Jin, Bo; Kim, Jungsik; Pi, Dong-Hai; Kim, Hyoung Seop; Meyyappan, M.; Lee, Jeong-Soo

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2014

In Situ Observation of Melting Behavior of ZnTe Nanowires

JOURNAL OF PHYSICAL CHEMISTRY C

Shaygan, Mehrdad; Gemming, Thomas; Bezugly, Viktor; Cuniberti, Gianaurelio; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 14

2014

Single-crystalline CdTe nanowire field effect transistors as nanowire-based photodetector

PHYSICAL CHEMISTRY CHEMICAL PHYSICS

Shaygan, Mehrdad; Davami, Keivan; Kheirabi, Nazli; Baek, Changi Ki; Cuniberti, Gianaurelio; Meyyappan, M.; Lee, Jeong-Soo

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2014

Thermal conductivity of ZnTe nanowires

JOURNAL OF APPLIED PHYSICS

Davami, Keivan; Weathers, Annie; Kheirabi, Nazli; Mortazavi, Bohayra; Pettes, Michael T.; Shi, Li; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 19

2013

Bandgap engineering of $\text{Cd}_x\text{Zn}_{1-x}\text{Te}$ nanowires

NANOSCALE

Davami, Keivan; Pohl, Judith; Shaygan, Mehrdad; Kheirabi, Nazli; Faryabi, Hamid; Cuniberti, Gianaurelio; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 9

2013

Reliability study of methods to suppress boron transient enhanced diffusion in high-k/metal gate Si/SiGe channel pMOSFETs

MICROELECTRONIC ENGINEERING

Park, Min Sang; Kim, Yonghyun; Lee, Kyong Taek; Kang, Chang Yong; Min, Byoung-Gi; Oh, Jungwoo; Majhi, Prashant; Tseng, Hsing-Huang; Lee, Jack C.; Banerjee, Sanjay K.; Lee, Jeong-Soo; Jammy, Raj; Jeong, Yoon-Ha

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2013

Improved Degradation and Recovery Characteristics of SiGe p-Channel Metal-Oxide-Semiconductor Field-Effect Transistors under Negative-Bias Temperature Stress

JAPANESE JOURNAL OF APPLIED PHYSICS

Choi, Do-Young; Sohn, Chang-Woo; Sagong, Hyun Chul; Jung, Eui-Young; Kang, Chang Yong; Lee, Jeong-Soo; Jeong, Yoon-Ha
被引用次数: 0 2013

Analytic Model of S/D Series Resistance in Trigate FinFETs With Polygonal Epitaxy

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sohn, Chang-Woo; Kang, Chang Yong; Ko, Myung-Dong; Choi, Do-Young; Sagong, Hyun Chul; Jeong, Eui-Young; Park, Chan-Hoon; Lee, Sang-Hyun; Kim, Ye-Ram; Baek, Chang-Ki; Lee, Jeong-Soo; Lee, Jack C.; Jeong, Yoon-Ha
被引用次数: 24 2013

Investigation of the electrical stability of Si-nanowire biologically sensitive field-effect transistors with embedded Ag/AgCl pseudo reference electrode

RSC ADVANCES

Rim, Taiuk; Kim, Kihyun; Hong, Nanki; Ko, Wooree; Baek, Chang-Ki; Jeon, Sangmin; Deen, M. Jamal; Meyyappan, M.; Jeong, Yoon-Ha; Lee, Jeong-Soo
被引用次数: 24 2013

Improved Electrical Characteristics of Honeycomb Nanowire ISFETs

IEEE ELECTRON DEVICE LETTERS

Rim, Taiuk; Kim, Kihyun; Kim, Sungho; Baek, Chang-Ki; Meyyappan, Meyya; Jeong, Yoon-Ha; Lee, Jeong-Soo
被引用次数: 23 2013

Simultaneous Formation of Ohmic Contacts on p +- and n +-4H-SiC Using a Ti/Ni Bilayer

JOURNAL OF ELECTRONIC MATERIALS

Joo, Sung-Jae; Baek, Sangwon; Kim, Sang-Cheol; Lee, Jeong-Soo
被引用次数: 16 2013

Simple S/D Series Resistance Extraction Method Optimized for Nanowire FETs

IEEE ELECTRON DEVICE LETTERS

Kim, Ye-Ram; Lee, Sang-Hyun; Sohn, Chang-Woo; Choi, Do-Young; Sagong, Hyun-Chul; Kim, Sungho; Jeong, Eui-Young; Kim, Dong-Won; Hong, Hyeongsun; Baek, Chang-Ki; Lee, Jeong-Soo; Jeong, Yoon-Ha
被引用次数: 8 2013

Improved performance of In₂Se₃ nanowire phase-change memory with SiO₂ passivation

SOLID-STATE ELECTRONICS

Baek, Chang-Ki; Kang, Daegun; Kim, JeongSik; Jin, Bo; Rim, Taiuk; Park, Sooyoung; Meyyappan, M.; Jeong, Yoon-Ha; Lee, Jeong-Soo
被引用次数: 19 2013

Investigation of electromigration in In₂Se₃ nanowire for phase change memory devices

APPLIED PHYSICS LETTERS

Kang, Daegun; Rim, Taiuk; Baek, Chang-Ki; Meyyappan, M.; Lee, Jeong-Soo
被引用次数: 16 2013

Thermally efficient and highly scalable In₂Se₃ nanowire phase change memory

JOURNAL OF APPLIED PHYSICS

Jin, Bo; Kang, Daegun; Kim, Jungsik; Meyyappan, M.; Lee, Jeong-Soo

被引用次数: 19

2013

Post-growth modification of electrical properties of ZnTe nanowires

CHEMICAL PHYSICS LETTERS

Faryabi, Hamid; Davami, Keivan; Kheirabi, Nazli; Shaygan, Mehrdad; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 6

2012

A computational and experimental investigation of the mechanical properties of single ZnTe nanowires

NANOSCALE

Davami, Keivan; Mortazavi, Bohayra; Ghassemi, Hessam M.; Yassar, Reza S.; Lee, Jeong-Soo; Remond, Yves; Meyyappan, M.

被引用次数: 24

2012

Investigation of Low-Frequency Noise Behavior After Hot-Carrier Stress in an n-Channel Junctionless Nanowire MOSFET

IEEE ELECTRON DEVICE LETTERS

Park, Chan-Hoon; Ko, Myung-Dong; Kim, Ki-Hyun; Lee, Sang-Hyun; Yoon, Jun-Sik; Lee, Jeong-Soo; Jeong, Yoon-Ha

被引用次数: 30

2012

Characterization of Channel-Diameter-Dependent Low-Frequency Noise in Silicon Nanowire Field-Effect Transistors

IEEE ELECTRON DEVICE LETTERS

Lee, Sang-Hyun; Baek, Chang-Ki; Park, Sooyoung; Kim, Dong-Won; Sohn, Dong Kyun; Lee, Jeong-Soo; Kim, Dae M.; Jeong, Yoon-Ha

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2012

Electron states in a silicon nanowire in the presence of surface potential and field

NANOTECHNOLOGY

Das, N. R.; Shinwari, M. W.; Deen, M. J.; Lee, Jeong-Soo

被引用次数: 3

2012

Device Design Guidelines for Nanoscale FinFETs in RF/Analog Applications

IEEE ELECTRON DEVICE LETTERS

Sohn, Chang-Woo; Kang, Chang Yong; Baek, Rock-Hyun; Choi, Do-Young; Sagong, Hyun Chul; Jeong, Eui-Young; Baek, Chang-Ki; Lee, Jeong-Soo; Lee, Jack C.; Jeong, Yoon-Ha

被引用次数: 13

2012

Thermal Breakdown of ZnTe Nanowires

CHEMPHYSICHEM

Davami, Keivan; Ghassemi, Hessam M.; Yassar, Reza S.; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 13

2012

Electrical characteristics of 20-nm junctionless Si nanowire transistors

SOLID-STATE ELECTRONICS

Park, Chan-Hoon; Ko, Myung-Dong; Kim, Ki-Hyun; Baek, Rock-Hyun; Sohn, Chang-Woo; Baek, Chang Ki; Park, Sooyoung; Deen, M. J.; Jeong, Yoon-Ha; Lee, Jeong-Soo

被引用次数: 133

2012

In situ observation of morphological change in CdTe nano- and submicron wires

NANOTECHNOLOGY

Davami, Keivan; Ghassemi, Hessem M.; Sun, Xuhui; Yassar, Reza S.; Lee, Jeong-Soo; Meyyappan, M.

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2011

Characterization and Modeling of 1/f Noise in Si-nanowire FETs: Effects of Cylindrical Geometry and Different Processing of Oxides

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Baek, Rock-Hyun; Baek, Chang-Ki; Choi, Hyun-Sik; Lee, Jeong-Soo; Yeoh, Yun Young; Yeo, Kyoung Hwan; Kim, Dong-Won; Kim, Kinam; Kim, Dae M.; Jeong, Yoon-Ha

被引用次数: 21

2011

Synthesis of ZnTe nanostructures by vapor-liquid-solid technique

CHEMICAL PHYSICS LETTERS

Devami, Keivan; Kang, Daegun; Lee, Jeong-Soo; Meyyappan, M.

被引用次数: 33

2011

Silicon nanowire ion sensitive field effect transistor with integrated Ag/AgCl electrode: pH sensing and noise characteristics

ANALYST

Kim, Sungho; Rim, Taiuk; Kim, Kihyun; Lee, Unsang; Baek, Eunhye; Lee, Hojoon; Baek, Chang-Ki; Meyyappan, M.; Deen, M. Jamal; Lee, Jeong-Soo

被引用次数: 77

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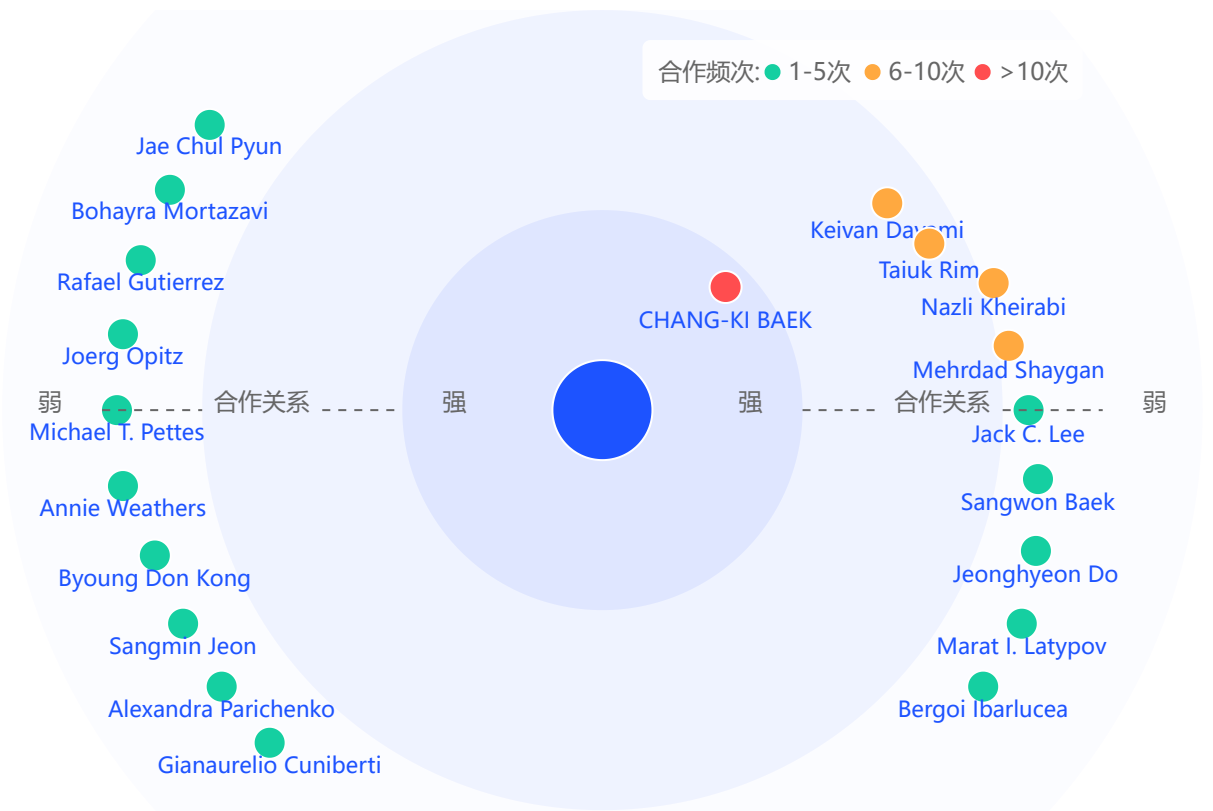
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